

CHE 2201 – Physical Chemistry II

Quantum Mechanics

Lec. 3. Mathematical Aspects of Quantum Mechanics

3.1. Information in a Wavefunction

In section 2.4, we have learned that a wavefunction contains all the dynamical information about the particle it describes (its location, momentum etc.). Further, the Born interpretation tells us as how to obtain the information from a wavefunction, about the location of a particle. So, the next question is how to extract the additional dynamical information from a wavefunction.

The time-independent Schrödinger equation for a particle of mass m moving in one dimension with energy E in a system is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \quad (3.1)$$

In addition, if we assume that if the potential energy is zero ($V=0$), then we can write

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \quad (3.2)$$

The solutions of the equation 3.2 will have the following form:

$$\psi = Ae^{ikx} + Be^{-ikx} \quad E = \frac{k^2\hbar^2}{2m} \quad (3.3)$$

where A and B are constants.

In order to confirm that ψ in equation 3.3 is a solution of equation 3.2, we have to substitute it into the left-hand side of the equation and show that we obtain $E\psi$.

$$\begin{aligned}
-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} &= -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} (Ae^{ikx} + Be^{-ikx}) \\
&= -\frac{\hbar^2}{2m} \{A(ik)^2 e^{ikx} + B(-ik)^2 e^{-ikx}\} \\
&= \frac{\hbar^2 k^2}{2m} (Ae^{ikx} + Be^{-ikx}) = E\psi
\end{aligned}$$

If B in equation 3.3 is equal to zero, then the wave function will be as follows:

$$\psi = Ae^{ikx} \quad (3.4)$$

To find out the whereabouts (the probability) of the particle, we have to calculate the probability density.

$$|\psi|^2 = (Ae^{ikx})^* (Ae^{ikx}) = (A^* e^{-ikx}) (Ae^{ikx}) = |A|^2 \quad (3.5)$$

This probability density is independent of x , and it means that wherever we look along the x -axis, there is an equal probability of finding the particle. In other words, if the wavefunction of the particle is given by equation 3.4, then we cannot predict where we will find it. The same would be true if A in equation 3.3 is equal to 0; then the probability density would be $|B|^2$ (is independent of x).

Now, let us assume that in the wavefunction $A = B$.

Then, the equation 3.3 becomes

$$\psi = A(e^{ikx} + e^{-ikx}) = 2A \cos kx \quad (3.6)$$

The probability density will now have the following form:

$$|\psi|^2 = (2A \cos kx)^* (2A \cos kx) = 4|A|^2 \cos^2 kx \quad (3.7)$$

This function is illustrated in Fig. 3.1.

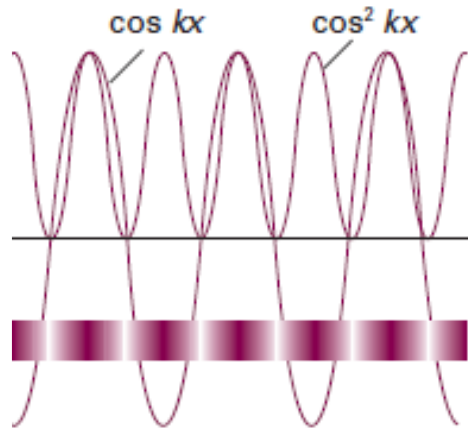


Fig. 3.1. Probability Distribution Corresponding to Eqn 3.7

As we can see, the probability density periodically varies between 0 and $4|A|^2$. The locations where the probability density is zero correspond to nodes in the wavefunction.

3.2. Operators, Eigenvalues and Eigenfunctions

In addition to wavefunctions, there is another set of mathematical objects that is of central importance in quantum mechanics, and they are called operators. The operators represent "observable quantities" such as position, momentum and energy; each observable has an operator associated with it. Operators "operate on" functions to give new functions, and hence their name.

$$\text{Operator} \times \text{Function} = \text{New Function} \quad (3.8)$$

An example of an operator is $\frac{d}{dx}$, and this operator says "differentiate with respect to x ".

Its effect on the function $\sin x$ is as follows:

$$\frac{d}{dx}(\sin x) = \cos x \quad (3.9),$$

so the new function is $\cos x$.

Operators can also be simple functions, for example, the operator x^2 means "multiply by x^2 ".

Here, it has to be noted that operators and functions cannot be reordered in the same way that numbers or functions can be.

For example:

2×3 is same as 3×2 ; and

$x \times \sin x$ is same as $\sin x \times x$.

But, $\frac{d}{dx} \times (\sin x)$ is not the same as $(\sin x) \times \frac{d}{dx}$.

As mentioned earlier, in general, operators act on functions to give another function as follows.

$$\text{Operator} \times \text{Function} = \text{New Function} \quad (3.8)$$

However, there are some functions, when acted upon by a given operator, are regenerated, but multiplied by a constant.

$$\text{Operator} \times \text{Function} = \text{Constant} \times \text{Function} \quad (3.10)$$

Such functions are said to be eigenfunctions of the operator and the constants are said to be the associated eigenvalues.

If the operator is $\hat{\Omega}$ (the hat is to distinguish it as an operator), then equation 3.10 can be written more formally as

$$\hat{\Omega} f_q = \omega f_q \quad (3.11)$$

where f_q is an eigenfunction of $\hat{\Omega}$ with eigenvalue ω . There may be more than one eigenfunction each with different eigenvalues. Equation 3.11 is known as the eigenvalue equation.

Example 1: Let us see whether $\exp(ax)$, where a is a constant, is an eigenfunction of the operator (d/dx) . In order to get the answer, we have to substitute the operator and function into the left-hand side of the eigenvalue equation (equation 3.11).

$$\left(\frac{d}{dx} \right) \exp(ax) = a \exp(ax) \quad (3.12)$$

So, from equation 3.12, we can see that operating on the function is to generate the original function multiplied by a constant. Therefore, we can say that $\exp(ax)$ is an eigenfunction of the operator (d/dx) with eigenvalue a .

Example 2: Is $\sin(ax)$, where a is a constant, is an eigenfunction of the operator (d/dx) ?

Let us substitute the operator and function into the left-hand side of the eigenvalue equation (equation 3.11).

$$\left(\frac{d}{dx}\right)\sin(ax) = a \cos(ax) \\ \neq \text{constant} \times \sin(ax) \quad (3.13)$$

Since the original function is not regenerated, we can say that $\sin(ax)$ is not an eigenfunction of the operator (d/dx) .

The time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \quad (3.1)$$

If the potential energy is zero ($V = 0$):

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi \quad (3.2)$$

Any version of Schrödinger equation may be written in the following concise form:

$$\hat{H}\psi = E\psi \quad (3.14)$$

where

$$\hat{H} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + V(x) \quad (3.15)$$

The quantity $\hat{H}$ is an operator, which is called the Hamiltonian Operator, named after the Irish physicist and mathematician William Hamilton (1805-1865), who developed a form of classical mechanics that was well suited to the formulation of quantum mechanics. In this case, the operation is to take the second derivative of ψ and (after multiplication by $-\hbar^2/2m$) to add the result to the outcome of multiplying ψ by V .

The operator $\hat{H}$ plays a special role in quantum mechanics, and it corresponds to the total energy of the system (i.e. the sum of the kinetic and potential energies). Thus, we can say

that the first term in equation 3.15 (the term proportional to the second derivative) must be the operator for the kinetic energy.

When we compare the Schrödinger equation in the form of 3.14 with equations 3.10 or 3.11, we can see that 3.14 is an eigenvalue equation.

If we replace the eigenfunction f_q in equation 3.11 by the wavefunction ψ , then we can write

$$\hat{\Omega}\psi = \omega\psi \quad (3.16)$$

Comparing the Schrödinger equation in the form of 3.14 with equation 3.16, we can conclude that the eigenfunction in equation 3.14 is the wavefunction corresponding to the energy E . So, when we are saying that “we have to solve the Schrödinger equation” means that “we have to find the eigenvalues and eigenfunctions of the Hamiltonian operator for the system”.

3.3. Construction of Operators

The importance of eigenvalue equations is that the pattern

$$(\text{Energy Operator})\psi = (\text{Energy})\times\psi \quad (3.17),$$

shown by the Schrödinger equation is repeated for other observables, or measurable properties of a system, such as the momentum or the electric dipole moment.

Thus, it is often the case that we can write

$$(\text{Operator corresponding to an Observable})\psi = (\text{Value of Observable}) \times \psi \quad (3.18)$$

Accordingly, the symbol $\hat{\Omega}$ in equation 3.16 could be interpreted as an operator (e.g. the Hamiltonian operator) corresponding to an observable (e.g. the energy), and the eigenvalue ω is the value of that observable (e.g. the value of the energy E).

So, if we know both the wavefunction ψ and the operator $\hat{\Omega}$ corresponding to the observable Ω of interest, and the wavefunction is an eigenfunction of the operator $\hat{\Omega}$, then we can predict the outcome of an observation of the property Ω (e.g. an atom's energy) by picking out the factor ω in the eigenvalue equation (equation 3.16).

A basic postulate of quantum mechanics tells us how to set up the operator corresponding to a given observable. Observables Ω , are represented by operators $\hat{\Omega}$, built from the following position and momentum operators.

$$\hat{x} = x \times \quad \hat{p}_x = \frac{\hbar}{i} \frac{d}{dx} \quad (3.19)$$

It means that the operator for location along the x -axis is multiplication (of the wavefunction) by x , and the operator for linear momentum parallel to the x -axis is proportional to taking the derivative (of the wavefunction) with respect to x .

Let us calculate linear momentum of a particle described by the wavefunction in equation 3.3, when (a). $B = 0$; and (b). $A = 0$.

$$\psi = Ae^{ikx} + Be^{-ikx} \quad (3.3)$$

Here, we operate on ψ with the operator corresponding to linear momentum and check the result. If the outcome is the original wavefunction multiplied by a constant (that is, we generate an eigenvalue equation), then the constant is identified as the value of the observable.

(a). In equation 3.3, $B = 0$

$$\hat{p}_x \psi = \frac{\hbar}{i} \frac{d\psi}{dx} = \frac{\hbar}{i} A \frac{de^{ikx}}{dx} = \frac{\hbar}{i} A \times ike^{ikx} = k\hbar Ae^{ikx} = k\hbar \psi \quad (3.20)$$

This is an eigenvalue equation, and by comparing it with equation 3.16, we find that

$$p_x = +k\hbar.$$

(b). In equation 3.3, $A = 0$

$$\hat{p}_x \psi = \frac{\hbar}{i} \frac{d\psi}{dx} = \frac{\hbar}{i} B \frac{de^{-ikx}}{dx} = \frac{\hbar}{i} B \times (-ik)e^{-ikx} = -k\hbar \psi \quad (3.21)$$

The magnitude of the linear momentum is the same in each case ($\hbar k$), but the signs are different. In (a) the particle is travelling to the right (positive x), but in (b) it is travelling to the left (negative x).

To construct operators for other spatial observables, we use the definitions in equation 3.19. For example, suppose we want to construct the operator for a potential energy of the form $V(x) = \frac{1}{2} kx^2$, where k is a constant. Then it follows from equation 3.19 that the operator corresponding to $V(x)$ is multiplication by x^2 .

$$\hat{V} = \frac{1}{2} kx^2 \times \quad (3.22)$$

To construct the operator for kinetic energy, we make use of the classical relation between kinetic energy and linear momentum, which in one dimension is $E_k = \frac{p_x^2}{2m}$.

Then by using the operator for p_x in equation 3.19, we find

$$\hat{E}_k = \frac{1}{2m} \left(\frac{\hbar}{i} \frac{d}{dx} \right) \left(\frac{\hbar}{i} \frac{d}{dx} \right) = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \quad (3.23)$$

It follows that the operator for the total energy (the Hamiltonian operator) is

$$\hat{H} = \hat{E}_k + \hat{V} = -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + \hat{V} \quad (3.24),$$

where $\hat{V}$ is the multiplicative operator in equation 3.22 (or some other appropriate expression for the potential energy).

The expression for the kinetic energy operator (equation 3.23), enables us to develop the interpretation of the Schrödinger equation. In mathematics, the second derivative of a function is a measure of its curvature: a large second derivative indicates a sharply curved function (Fig. 3.2).

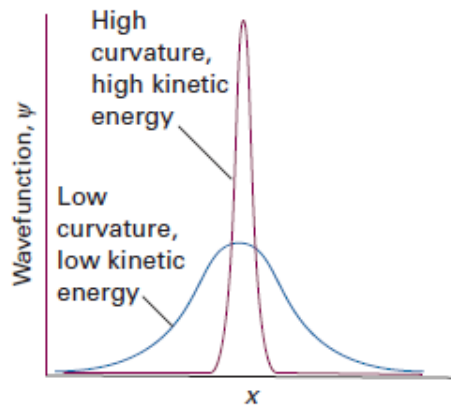


Fig. 3.2. Curvature of the Wavefunction and Kinetic Energy

It follows that a sharply curved wavefunction (ψ) is associated with a high kinetic energy, and one with a low curvature is associated with a low kinetic energy. This interpretation is in line with the de Broglie equation (refer section 2.2), which predicts a short wavelength (a sharply curved wavefunction) when the linear momentum (and hence the kinetic energy) is high.

The curvature of a ψ in general varies from place to place. Wherever a ψ is sharply curved, its contribution to the total kinetic energy is large. Wherever the ψ is not sharply curved, its contribution to the overall kinetic energy is low.

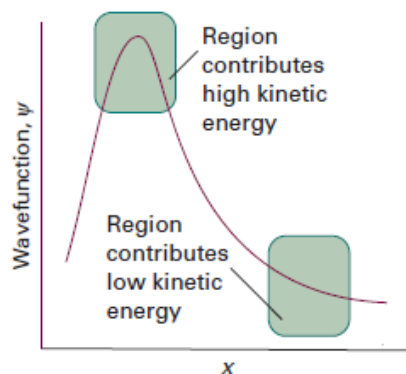


Fig. 3.3. Contribution of Different Regions of ψ to Kinetic Energy

The observed kinetic energy of the particle is an integral of all the contributions of the kinetic energy from each region. Hence, we can expect a particle to have a high kinetic energy if the average curvature of its' ψ is high. Locally there can be both positive and

negative contributions to the kinetic energy (because the curvature can be either positive or negative), but the average is always positive.

The association of high curvature with high kinetic energy will turn out to be a valuable guide to the interpretation of ψ s and the prediction of their shapes. For example, suppose we need to know the ψ of a particle with a given total energy and a potential energy that decreases with increasing x (Fig. 3.4).

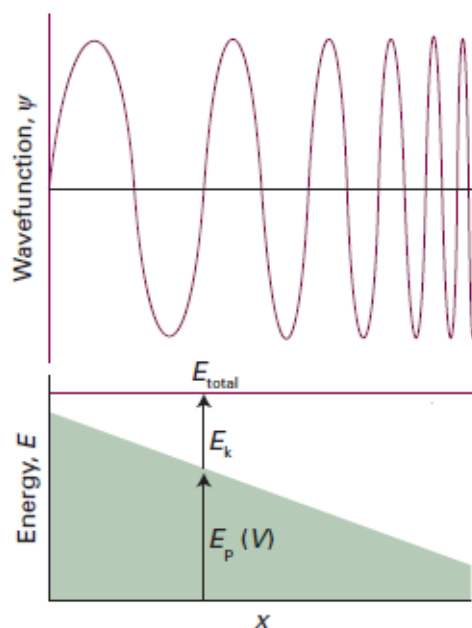


Fig. 3.4. ψ of a Particle, with Potential Energy decreases Towards Right
(Only the real part of ψ is shown, the imaginary part is similar but displaced to the right)

Because the difference $E - V = E_k$ increases from left to right, the ψ must become more sharply curved as x increases: its wavelength decreases as the local contributions to its kinetic energy increase. We can therefore guess that the ψ will look like the function depicted in Fig. 3.4, and more detailed calculation confirms this to be so.

3.4. Hermitian Operators

All the quantum mechanical operators that correspond to observables have a very special mathematical property, i.e. they are 'Hermitian'.

A Hermitian operator is one for which the following relation is true:

$$\text{Hermiticity: } \int \psi_i^* \hat{Q} \psi_j d\tau = \left\{ \int \psi_j^* \hat{Q} \psi_i d\tau \right\}^* \quad (3.25)$$

where $d\tau = dx dy dz$.

That is, the same result is obtained by letting the operator act on ψ_j and then integrating or by letting it act on ψ_i instead, integrating, and then taking the complex conjugate of the result. One consequence of hermiticity is that it reduces the number of integrals we need to evaluate.

It is easy to confirm that the position operator (x) is Hermitian because we are free to change the order of the factors in the following integral:

$$\int_{-\infty}^{\infty} \psi_i^* x \psi_j d\tau = \int_{-\infty}^{\infty} \psi_j x \psi_i^* d\tau = \left\{ \int_{-\infty}^{\infty} \psi_j^* x \psi_i d\tau \right\}^* \quad (3.26)$$

Hermitian operators are extremely important due to two properties: their eigenvalues are real and their eigenfunctions are ‘orthogonal’.

All observables have real values (in the mathematical sense, such as $x = 2$ m and $E = 10$ J), so all observables are represented by Hermitian operators.

To say that two different functions ψ_i and ψ_j are orthogonal means that the integral (over all space) of their product is zero:

$$\int \psi_i^* \psi_j d\tau = 0 \quad \text{for} \quad i \neq j \quad (3.27)$$

A general feature of quantum mechanics is that wavefunctions corresponding to different eigenvalues of a Hermitian operator are orthogonal. For example, the Hamiltonian operator is Hermitian (it corresponds to an observable, the energy). Therefore, if ψ_1 corresponds to one energy, and ψ_2 corresponds to a different energy, then we know at once that the two functions are orthogonal and that the integral of their product is zero.

The property of orthogonality is very important in quantum mechanics because it enables us to eliminate a large number of integrals from calculations. Orthogonality plays a

central role in the theory of chemical bonding and spectroscopy. Sets of functions that are normalized and mutually orthogonal are called orthonormal.

3.5. Expectation Values

Suppose that the ψ is given by the equation 3.3:

$$\psi = Ae^{ikx} + Be^{-ikx} \quad (3.3)$$

If $A = B$, then

$$\psi = A(e^{ikx} + e^{-ikx}) = 2A \cos kx \quad (3.6)$$

In order to find the linear momentum described by this wavefunction, we have to use the definition in equation 3.19. Then we get

$$\hat{p}_x \psi = \frac{\hbar}{i} \frac{d\psi}{dx} = \frac{2\hbar}{i} A \frac{d \cos kx}{dx} = -\frac{2k\hbar}{i} A \sin kx \quad (3.28)$$

This expression is not an eigenvalue equation, because the function on the right ($\sin kx$) is different from that on the left ($\cos kx$).

When the ψ of a particle is not an eigenfunction of an operator, the property to which the operator corresponds does not have a definite value. However, in the current example the momentum is not completely indefinite because the cosine wavefunction is a linear combination of e^{ikx} and e^{-ikx} , and these two functions, individually corresponds to definite momentum states. We say that the total wavefunction is a superposition of more than one wavefunction.

Symbolically we can write the superposition as

$$\psi = \underbrace{\psi_{\rightarrow}}_{\substack{\text{Particle with} \\ \text{linear} \\ \text{momentum} \\ +\hbar k}} + \underbrace{\psi_{\leftarrow}}_{\substack{\text{Particle with} \\ \text{linear} \\ \text{momentum} \\ -\hbar k}}$$

[Note: In general, a linear combination of two functions f and g is $(C_1 f + C_2 g)$, where C_1 and C_2 are numerical coefficients, so a linear combination is a more general term than 'sum'. In a sum, $C_1 = C_2$

= 1. A linear combination might have the form, for example $(0.567f + 1.234g)$, so it is more general than the simple sum $f + g$.]

The interpretation of this composite wavefunction is that, if the momentum of the particle is repeatedly measured in a long series of observations, then its magnitude will be found to be $\hbar k$ in all the measurements (because that is the value for each component of the wavefunction). However, because the two component wavefunctions occur equally in the superposition, half the measurements will show that the particle is moving to the right ($P_x = +\hbar k$), and half the measurements will show that it is moving to the left ($P_x = -\hbar k$). According to quantum mechanics, we cannot predict in which direction the particle will be travelling. All we can say is that, in a long series of observations, if the particle is described by this wavefunction then there are equal probabilities of finding the particle travelling to the right and to the left.

The same interpretation applies to any wavefunction written as a linear combination of eigenfunctions of an operator. Thus, suppose the wavefunction is known to be a superposition of many different linear momentum eigenfunctions and written as the linear combination

$$\psi = c_1\psi_1 + c_2\psi_2 + \cdots = \sum_k c_k\psi_k \quad (3.29)$$

where the C_k are numerical coefficients and the ψ_k correspond to different momentum states. The functions ψ_k are said to form a complete set in the sense that any arbitrary function can be expressed as a linear combination of them.

So, according to quantum mechanics:

1. When the momentum is measured, in a single observation one of the eigenvalues corresponding to the ψ_k that contribute to the superposition will be found.
2. The probability of measuring a particular eigenvalue in a series of observations is proportional to the square modulus ($|C_k|^2$) of the corresponding coefficient in the linear combination.
3. The average value of a large number of observations is given by the expectation value, $\langle\Omega\rangle$, of the operator corresponding to the observable of interest.

The expectation value of an operator $\hat{\Omega}$ is defined as

$$\langle \Omega \rangle = \int \psi^* \hat{\Omega} \psi d\tau \quad (3.30)$$

This formula is valid only for normalized wavefunctions. Thus, an expectation value is the weighted average of a large number of observations of a property.

If ψ is an eigenfunction of $\hat{\Omega}$ with eigenvalue ω , the expectation value of $\langle \Omega \rangle$ is

$$\langle \Omega \rangle = \int \psi^* \hat{\Omega} \psi d\tau = \int \psi^* \omega \psi d\tau = \omega \int \psi^* \psi d\tau = \omega \quad (3.31)$$

Since ω is a constant and may be taken outside the integral, and the resulting integral is equal to 1 for a normalized wavefunction. The interpretation of this expression is that, because every observation of the property Ω results in the value ω (because the wavefunction is an eigenfunction of $\hat{\Omega}$), the mean value of all the observations is also ω .

3.6. Postulates of Quantum Mechanics – A Summary

Let us summarise the postulates on which quantum mechanics is based and which have been discussed during this lecture.

Wavefunction: All dynamical information is contained in the wavefunction ψ for the system, which is a mathematical function found by solving the Schrödinger equation for the system.

In one dimension:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \quad (3.1)$$

Born Interpretation: If the wavefunction of a particle has the value ψ at some point x , then the probability of finding the particle in an infinitesimal volume $d\tau = dx dy dz$ at that point is proportional to $|\psi|^2 d\tau$.

Acceptable Wavefunctions: An acceptable wavefunction must be continuous, have a continuous first derivative, be single-valued, and be square-integrable.

Observables: Observables Ω , are represented by operators $\hat{\Omega}$, built from the following position and momentum operators:

$$\hat{x} = x \times \quad \hat{p}_x = \frac{\hbar}{i} \frac{d}{dx} \quad (3.19)$$

In addition, (in lecture 2) we have learned 3 basic principles of quantum mechanics, i.e. quantization of energy, wave-particle duality and the uncertainty principle.